

PopSignAI: Integrating Sign Recognition into Gameplay to Teach Sign Language

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Abstract

With approximately 95% of deaf children born to hearing parents, the need for accessible, self-paced, and effective sign language learning tools is critical. This paper explores the integration of Sign Language Recognition (SLR) models into educational games for teaching American Sign Language (ASL) production, focusing on the development of PopSignAI, an arcade-style bubble-bursting game designed to teach ASL sign production through gesture recognition. PopSignAI builds on the existing PopSign game, which teaches sign reception, by incorporating a 560-sign SLR model to recognize player-generated signs. Through the development of this game, we demonstrate the potential of integrating SLR models into sign language game mechanics as an effective method for teaching sign production. We also outline key considerations for their integration, addressing challenges related to cognitive load, gameplay intuitiveness, and the technical demands of gesture recognition.

CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods**; *Interaction design process and methods*; *Ubiquitous and mobile computing systems and tools*; *Gestural input*; • **Applied computing** → **E-learning**.

Keywords

Sign Language Recognition, ASL, Educational Games, Accessibility

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1 Introduction

Approximately 95% of deaf infants are born to hearing parents who are often unfamiliar with American Sign Language (ASL) [19, 20]. Lack of early exposure to a naturally accessible language during critical language development periods puts these children at risk of Language Deprivation Syndrome (LDS) [11, 13]. Despite these risks, parents of deaf or hard of hearing children can face a difficult time learning ASL to communicate with their children, due to difficulties with language acquisition later in life, and lack of time required to take regular ASL lessons [27]. Thus, being able to learn ASL in an easy, self-paced, and ubiquitous manner is highly desirable to such audiences.

To address this need, we created PopSignAI, an arcade-style bubble-bursting game for learning sign production, in which players are required to perform ASL signs in order to blow bubbles out of a cannon. PopSignAI builds on the existing gameplay of PopSign, a game originally targeted at teaching sign recognition, and pivots it towards sign production by integrating into it our 560-sign ASL recognizer—an ML model capable of identifying signs based on gesture input. In this paper we present the updated game mechanics of PopSignAI, and we argue that the introduction of ML-based gesture recognition as a mode of input for sign language games introduces significant complexities to game interactions, but, when implemented thoughtfully, can be an effective way for users to learn sign production.

2 Background and Related Work

2.1 Sign Language Games

Educational games have been widely studied and recognized as an effective tool for language acquisition [1, 15, 25, 27, 28]. However, the subset of games designed specifically for teaching sign language is much smaller and comes with limitations. A considerable number of these games lack integrated sign recognizers, like PopSign [12], Spell Well [9], ASL Tales and Games for Kids [7], SmartSignPlay [3], Con-Sign-Tration [8], Sign-O ASL Bingo [5], and Rosie's Walk [22]. While these games are effective at helping learners recognize ASL signs using methods such as multiple-choice questions, video demonstrations, and similar mechanics, they fall short in teaching users how to produce signs.

The existing games for learning sign production incorporate sign recognizers to do so, but come with limitations. For example, games like SignFind [21] and iSpy-uSpy [2] are able to recognize signs but have vocabularies restricted to 10 signs and 6 signs respectively, due to limitations of the recognition systems. Games like KinectSign [17], Fingerspelling.xyz [6], MatLIBRAS [24], Hangman [4], and Didactic Games [23] also integrate sign recognizers, but they are limited to identifying fingerspelling rather than full words.

2.2 Sign Language Recognition in Apps

In this paper, we use Sign Language Recognition (SLR) synonymously with Isolated Sign Language Recognition (ISLR). ISLR focuses on individual sign recognition whereas Continuous Sign Language Recognition (CSLR) focuses on real-time translations of sign language. Both ISLR and CSLR are part of SLR but in this paper we only focus on ISLR.

There have been long standing constraints on developments in SLR apps due to the limitations of real-time hand-gesture recognition, such as its resource-intensive nature, its dependence on 3D camera input (difficult on a phone) [16, 29], and the lack of annotated training data to build accurate and efficient ML models for SLR. However, the recent availability of large annotated sign language datasets through PopSign ASL v1.0 [26] and through our data collection application, RecordTheseHands¹, coupled with the recent advances in hand-tracking and model inference software [18] makes efficient sign recognition on smartphones an achievable goal. These advancements mark a significant step toward improving gesture input and facilitating sign language learning on mobile devices [10, 14], paving the way for the development of new sign language applications.

Leveraging the largest publicly available dataset of isolated signs [26] along with data from RecordTheseHands, we have created a sign language recognition model that is capable of collecting pose input from a user's phone camera. Upon a trigger, the data is passed to a TFLite model which classifies the data as a particular sign from its 560-sign vocabulary. Our goal was to design gameplay that incorporates this recognizer as a core mechanic, with the ultimate goal of teaching sign production.

3 Original PopSign for Teaching Sign Reception

To investigate the implications of integrating sign recognition into conventional game mechanics, we decided to build upon an existing ASL app targeted at teaching sign reception, PopSign (Fig. 1).

PopSign is a classic arcade-style bubble-bursting game. Players begin each level by watching videos of five unique ASL vocabulary words. They are then presented with a set of bubbles, each of which is tagged with one of the previously learned words. A video is played which depicts a person signing the word associated with the next bubble that is loaded into the cannon. The player must recognize the sign from the video to identify the loaded bubble, and aim and release the bubble accordingly. As in traditional bubble-bursting games, grouping three or more bubbles of the same kind causes them to pop. The level is complete when all bubbles are cleared. The gameplay mechanics also follow those similar to existing bubble

bursting games, wherein the player drags their finger to aim the cannon, and releases their finger to confirm their shot.

By requiring players to recognize and identify signs to direct a bubble, the game reinforces their sign recall through repeated practice, making it a highly effective method for teaching sign reception. However, the main drawback of PopSign is that it does not teach sign production.

4 Adding Sign Recognition in PopSignAI for Teaching Sign Production

Developing the ability to produce words is a crucial step toward achieving proficiency in any language, and ASL is no exception. Parents of deaf children have shown a desire to be able to practice sign production in order to build confidence when signing to their children or other deaf adults [12]. We contend that an integrated sign recognizer capable of detecting the learner's produced signs and providing immediate feedback would address this critical need.

Thus, we created PopSignAI (Fig. 2), an arcade-style bubble bursting game which functions very similarly to PopSign, but is targeted at teaching sign production through our integrated SLR model. The main difference is that players are able to self-select what bubble they want shot out of the cannon by signing the word for that particular bubble. The SLR model identifies the signed word, and the corresponding bubble is launched from the cannon. We also implemented a signifier to indicate to players that the recognizer is actively detecting their hand, in order to increase user confidence in the detection capabilities of the recognizer itself. This feedback is provided through a skeleton-like overlay that appears on the user's hand during recording (Fig. 2), signaling that the camera is successfully tracking their movements.

There were numerous considerations involved in designing the gameplay mechanics for PopSignAI's cannon-shooting mechanism. The SLR model requires users to hold their phones at arm's length when signing in order to align with the videos from the training data, necessitating gameplay that encourages this posture without explicit instructions. Since the SLR model cannot run continuously due to power consumption and false positives, an in-game trigger is needed to signal the model to analyze the user's video at the right moment. This trigger needs to be user-initiated to provide them with sufficient control over gameplay, and has to be activated just before or after signing, ensuring the SLR model captures enough frames to accurately classify the signs. Additionally, the act of signing needs to feel intuitive and seamlessly integrated into gameplay to minimize cognitive load on the player and maintain their focus on learning ASL.

A key decision was whether to design for one-handed or two-handed play. To achieve one-handed play, we separated aiming and shooting into distinct actions, replacing the previous 'drag to aim and release to shoot' mechanism with a setup where a dedicated 'shoot' button acts as a trigger for the SLR model. This design requires the player to aim the cannon, perform the sign, and press the 'shoot' button in a successive flow, all using their dominant hand, while holding the phone at arm's length with their non-dominant hand. This 'shoot' button triggers the recognizer, which identifies the sign and the cannon releases the corresponding bubble.

¹<https://github.com/Accessible-Technology-in-Sign/RecordTheseHands>

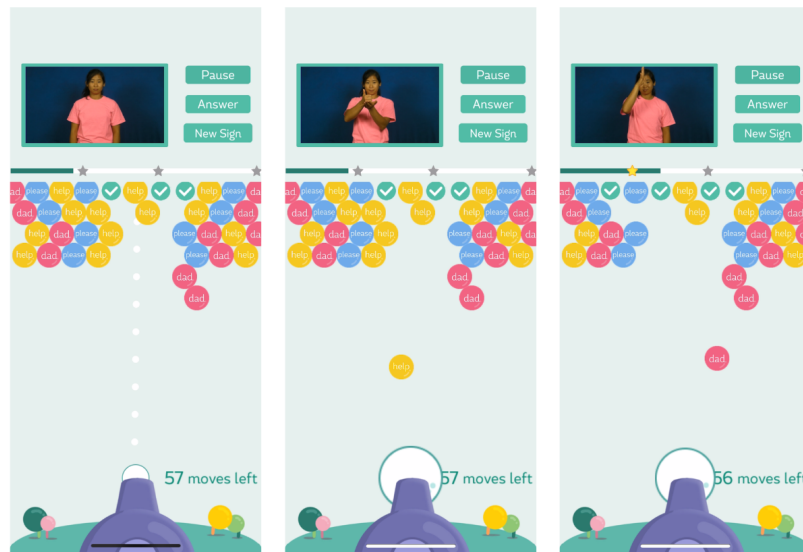


Figure 1: PopSignAI gameplay to teach sign reception

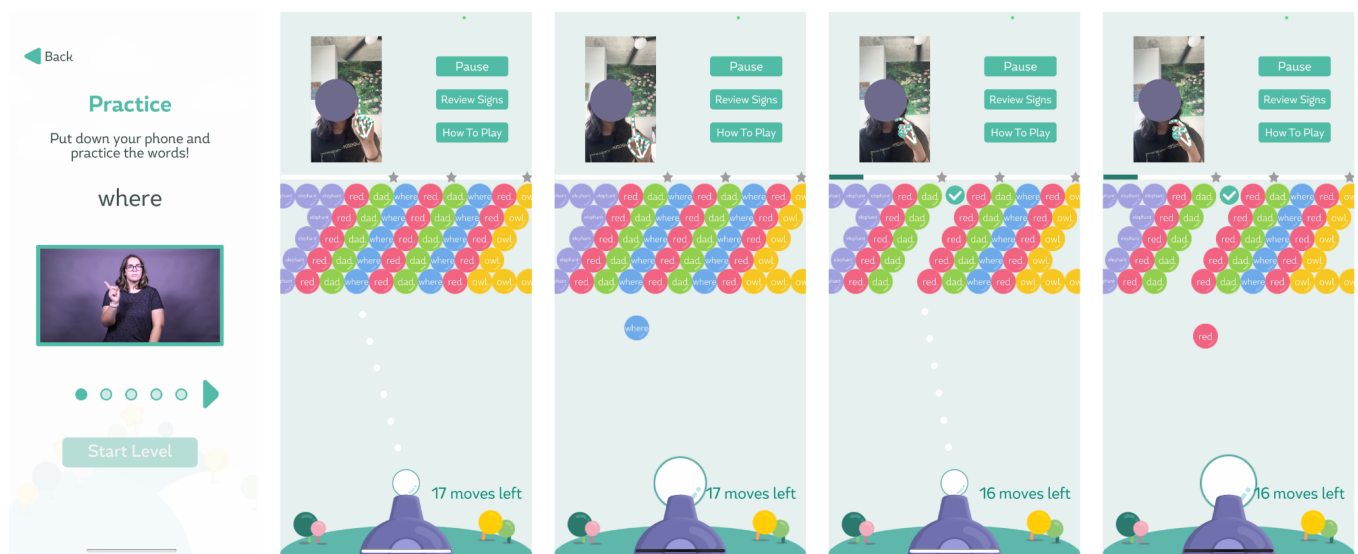


Figure 2: PopSignAI gameplay to teach sign production

This approach improves accessibility for players with dexterity challenges that make two-handed play difficult.

However, this mode of one-handed play presents several drawbacks. Players often brought the phone closer to their face while aiming, causing improper framing during signing or requiring awkward back-and-forth repositioning. Also, the reliance on a ‘shoot’ button to confirm signs introduced delays, as the system only processed the last 60 frames, risking missed recognition if the sign wasn’t fully captured.

In two-handed play, the player signs with one hand while using the other to aim and shoot, resolving these issues. Firstly, it minimizes improper framing, as each hand is assigned specific and consistent actions, reducing the likelihood of the player changing the placement of their hands between tasks. Secondly, it eliminates delays caused by late triggering of the recognizer, as we no longer have the need for a separate 'shoot' button to act as a recognizer trigger. The player is now able to perform the action of aiming and signing in parallel, and so we can use this affordance to integrate

the recognizer trigger with the aiming action. Holding their finger down to aim prompts the player to sign, and releasing it confirms the aim, triggers the recognizer, and launches the corresponding bubble. For these reasons, the two-handed model was ultimately chosen.

5 Critical Reflection

The development of PopSignAI represents a significant step forward in the integration of sign recognition technology within educational gaming, specifically for teaching sign production. Given the demand and educational effectiveness of self-paced games that teach sign language, it is crucial to explore scalable approaches for their development. One of the biggest roadblocks to doing so is the complexity of designing game mechanics that effectively integrate an SLR model. Through our work integrating an SLR model into an existing sign language game, we have identified several key considerations essential to achieving this balance.

Firstly, the design should allow for two-handed play, to ensure that the player is able to dedicate one hand specifically for the task of signing, and the other hand for the rest of the gameplay. This prevents situations in which the player moves the phone closer for gameplay and further away for signing, which could result in misalignment with the camera and lead to failures in the SLR model's sign detection. Secondly, the design should include an in-game, user-initiated trigger for the SLR to begin sign identification, either before or directly after the player begins signing. Since the SLR model cannot run continuously without generating errors or null cases, this trigger ensures accurate and timely recognition. Lastly, the design should ensure that the act of signing is not disjointed from the gameplay itself, in order to create an organic and intuitive experience for the player.

There are limitations in the two-handed model, such as its reliance on dexterity in both hands, which may exclude players with motor impairments or limited hand mobility. The above outlined considerations also inherently limit the types of games that can be created, as gameplay must prioritize compatibility with SLR models and signing mechanics over other creative or complex interactions, placing constraints on game design and creativity in sign language games.

6 Conclusion and Future Work

In conclusion, this study outlines a set of key considerations for integrating SLR models into sign language games aimed at teaching both ASL reception and production, while demonstrating the effectiveness of this approach as a game mechanic. Future work will test the SLR accuracy of various new game mechanics and will also include the release of a SLR toolkit for popular game development platforms such as Unity, iOS, and Android, to simplify the technical and gameplay-related complexities associated with SLR integration in games. Through this work, we hope to take a significant step toward democratizing the development of sign language learning tools, empowering a broader community of developers to create scalable and impactful educational games.

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